SPGFormer: Structure Perception Graph Transformer With Laplacian Position Encoding for Hyperspectral Image Classification

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***Abstract*—With the integration of graph structure representation and self-attention mechanism, the graph Transformer (GT) demonstrates remarkable effectiveness in hyperspectral image (HSI) classification by simultaneously modeling complex topological structures and capturing long-range spectral–spatial dependencies. Current GT models struggle to effectively integrate node features with position information while balancing global positional context and local structural relationships. To address these challenges, this article proposes a structure perception GT (SPGFormer) with Laplacian position encoding (PE) for HSI classification. Specifically, SPGFormer introduces a dual-branch interactive Transformer (DBIT) module that processes node spectral features and Laplacian position information in parallel by the node feature Transformer (NFT) branch and the structure perception position Transformer (SPPT) branch, respectively. The NFT branch employs a Transformer encoder to model spectral characteristics, while the SPPT branch leverages graph Laplacian eigenvectors to capture global position information. In addition, a structure perception multihead self-attention (SPMHSA) mechanism is proposed to integrate the local topological structural information into the SPPT branch. Moreover, to facilitate effective interaction between these complementary feature spaces, a bidirectional cross-attention module (BCAM) is proposed to enable bidirectional communication between these two branches. Comprehensive experimental results demonstrate that SPGFormer outperforms existing state-of-the-art (SOTA) methods in multiple metrics, including overall accuracy (OA), average accuracy (AA), and Kappa coefficient, with improvements in OA of 2.04%, 2.39%, 1.10%, and 2.27%, respectively, on four benchmark HSI datasets.**

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***Index Terms*—Cross attention, graph Laplacian position encoding (PE), hyperspectral image (HSI) classification, Transformer.**

NOMENCLATURE

Structure Perception Position Transformer (SPPT) Branch

$\mathbf{H}_{\text{str}}^{(i)}$	Input features of SPPT branch at layer i.
$\mathbf{Q}_{\text{str}}^{(i)}$, $\mathbf{K}_{\text{str}}^{(i)}$, and $\mathbf{V}_{\text{str}}^{(i)}$	$\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ matrices of SPPT branch at layer i.
$\mathbf{W}_{Q,\text{str}}^{(i)}$, $\mathbf{W}_{K,\text{str}}^{(i)}$, and $\mathbf{W}_{V,\text{str}}^{(i)}$	Projection weights for $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ of SPPT branch at layer i.
$\mathbf{A}_{\text{str}}^{(i)}$	Multihead self-attention output of SPPT branch at layer i.
$\mathbf{O}_{\text{str}}^{(i)}$	Final output of SPPT branch at layer i.

Node Feature Transformer (NFT) Branch

$\mathbf{H}_{\text{fea}}^{(i)}$	Input features of NFT branch at layer i.
$\mathbf{Q}_{\text{fea}}^{(i)}$, $\mathbf{K}_{\text{fea}}^{(i)}$, and $\mathbf{V}_{\text{fea}}^{(i)}$	$\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ matrices of NFT branch at layer i.
$\mathbf{W}_{Q,\text{fea}}^{(i)}$, $\mathbf{W}_{K,\text{fea}}^{(i)}$, and $\mathbf{W}_{V,\text{fea}}^{(i)}$	Projection weights for $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ of NFT branch at layer i.
$\mathbf{A}_{\text{fea}}^{(i)}$	Multihead self-attention output of NFT branch at layer i.
$\mathbf{O}_{\text{fea}}^{(i)}$	Final output of NFT branch at layer i.

Bidirectional Cross-Attention Module (BCAM)

$\mathbf{W}_{\text{Q,cro}}$, $\mathbf{W}_{\text{K,cro}}$, and $\mathbf{W}_{\text{V,cro}}$	Cross-attention projection matrices.
$\hat{\mathbf{O}}_{\text{str}}^{(i)}$ and $\hat{\mathbf{O}}_{\text{fea}}^{(i)}$	Branch outputs after BCAM at layer i.

Feature Fusion

$\mathbf{H}_{\text{str}}^{(i+1)}$	Output of the SPPT branch at layer $(i+1)$.
$\mathbf{H}_{\text{fea}}^{(i+1)}$	Output of the NFT branch at layer $(i+1)$.

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$\mathbf{H}_{\text{trs}}$	Fused features from SPPT and NFT branches.
$\mathbf{H}_{\text{cnn}}$	Pixel-level features from convolutional neural network (CNN) branch.
$\mathbf{H}_{\text{fus}}$	Final fused features.
α	Learnable fusion weight parameter.

I. Introduction

THE rich spectral information make hyperspectral image (HSI) valuable for numerous applications, including land-cover classification, precision agriculture, urban planning, environmental monitoring, and resource detection. HSI classification technology is a fundamental task in HSI processing. In recent years, deep learning methods, such as CNNs, residual networks (ResNets), long short-term memory (LSTM) networks, and generative adversarial networks (GANs), have been widely applied to HSI classification with impressive performance. Despite their excellent performance in feature extraction, these methods often fail to fully capture long-range dependencies and non-Euclidean structural relationships in HSI data due to their inherent limitations.

The distribution of land covers in HSI exhibits significant spatial correlation, which can be naturally and effectively expressed and analyzed through graph structures. However, graph-based analytical frameworks primarily focus on aggregating features from adjacent nodes, often neglecting the importance of long-distance dependencies. In recent years, Transformer has demonstrated exceptional advantages in capturing global dependencies through their powerful multihead self-attention mechanisms, achieving breakthrough advances in natural language processing, computer vision, and other fields. The application of Transformers to graph-structured data, known as graph Transformers (GTs), has become a research hotspot and yielded satisfactory results in HSI processing [1], [2]. Spatial–spectral features enhanced graphormer (S^2GFormer) embeds graph convolution into Transformers through a follow patch mechanism, achieving effective fusion of local and global features [3]. Pyramid superpixel transformer (PSFormer) adopts a progressive superpixel merging strategy, implementing multiscale feature extraction through pyramid structure design [4].

In Transformer architectures, position encoding (PE) is crucial to understanding the sequential and spatial relationships between input elements [5]. Unlike the simple 1-D or 2-D positional relationships in sequential data or regular grid data, graph-structured data present complex and diverse topological characteristics, including varying node degrees, nonuniform edge weights, and diverse connection patterns. When extending PE concepts from sequences and regular grids to graph-structured data, the first challenge is how to model the position information. Initially, basic PEs, such as random PE [6], absolute PE [7], [8], and relative PE [9], [10], were first applied to sequential data and regular grid data, before being extended to graph-structured data processing. Based on these foundations, researchers proposed rotary PE [11] with rotational invariance, which encodes positional information through rotation operations in the complex plane, significantly improving the robustness to positional transformations. To further improve encoding capabilities, composite PE strategies, such as double PE [12] and 3-D PE [13], provide richer positional representations by combining multiple types of positional information. Recently, research focus has shifted to PE methods specifically designed for graph structural characteristics, such as rotation-invariant PE (RIPE) [14], advanced absolute PE [15], and Laplacian PE [16], that directly reflect the topological structure of the graph.

Another issue in GT PE is how to effectively integrate PE with node features. Current fusion strategies primarily follow two approaches: direct addition of PE to node features or integration as bias terms in attention calculations. Although direct addition has proven effective for sequence and grid data, node features with richer semantic content and larger magnitudes often overshadow positional information, particularly as network depth increases, limiting the model's ability to leverage graph topology. The bias term approach partially preserves positional information independence by directly influencing attention distribution, yet still fails to maintain complete independence. In hyperspectral graph models, positional information encode both spatial adjacency and spectral similarity, critical factors for distinguishing land-cover types with similar spectral signatures but different spatial distributions. Consequently, treating positional information as merely supplementary to node features or as an attention bias term is insufficient for optimal performance.

Despite significant advances in GT for HSI classification, there are some challenges that hold the following.

1) Existing GT models typically input node features and PE into the network after simple fusion, causing positional information to be easily dominated by node features, preventing full retention and utilization of graph structure information.
2) The available PE methods face significant challenges in effectively encoding complex topological information, especially in modeling the global positional information and local structural details within the graph.
3) Existing methods neglect the interactive relationship between node features and positional information, which prevents full utilization of the complementary information contained in these two essential representations.

Motivated by the analysis above, we propose a structure perception GT (SPGFormer) model. The core design philosophy of this model is to achieve balanced expression of positional information and node features, as well as coordinated modeling of global position and local structure through a parallel dual-branch architecture and structure perception attention mechanism. Specifically, we first apply the simple linear iterative clustering (SLIC) [17] to perform superpixel segmentation on HSI and construct a graph model based on spatial adjacency relationships and spectral similarity. The spectral features of superpixel nodes and graph Laplacian matrix eigenvectors serve as two independent Transformer input branches, building parallel feature encoding and structure encoding pathways. Simultaneously, we incorporate k-hop local structural information into the positional Transformer branch, proposing

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a structure perception multihead self-attention (SPMHSA). This enables the model to simultaneously attend both global positions and local connection patterns of nodes. Furthermore, to facilitate effective interaction between spectral features and positional structural features, we designed a BCAM that enables bidirectional information flow between the two branches. The main innovations of this article are as follows.

1) We designed a dual-path Transformer architecture that processes the spectral features of superpixel nodes and graph Laplacian eigenvectors as two independent input branches in parallel.
2) We proposed a structure-aware multihead attention mechanism that incorporates k-hop local structural information into the positional Transformer branch as bias terms. This mechanism enables the model to simultaneously capture global positional relationships and local connection patterns.
3) We designed a BCAM that enables bidirectional interaction and enhancement between node features and position information, which promotes effective fusion of both types of information, achieving mutual guidance between node features and position information.

The remainder of this article is organized as follows. Section II systematically reviews related work. Section III presents the proposed SPGFormer model in detail. Section IV demonstrates experimental results, and Section V concludes this article and discusses directions for future research.

II. Related Works

A. GT for HSI Classification

Graph-based methods have achieved remarkable success in HSI processing due to their ability to model complex spatial–spectral relationships and handle non-Euclidean structures inherent in HSI. Various graph-based approaches have been widely applied and demonstrated significant success in HSI analysis [18], [19], [20]. In recent years, Transformer architectures have also become a research hotspot in the hyperspectral domain due to their exceptional capability in capturing long-range dependencies and modeling global contextual relationships through powerful self-attention mechanisms [21]. Building upon these advances, the integration of graph structures with Transformer architectures has led to the emergence of GTs as a powerful approach for effectively capturing both global topological relationships and long-range dependencies in HSI. GT successfully overcomes the limitations of traditional CNN models, which suffer from restricted receptive fields, as well as the over-smoothing problems encountered by GCNs when processing long-distance node dependencies. In addition, the GT avoids the inherent limitations of standard Transformer architectures that neglect the spatial topological structure [22]. Furthermore, the GT demonstrates significant advantages in establishing long-range dependencies between spectral bands while preserving the essential topological structure of the data. This capability enables a more effective balance between local spatial features and global semantic information, resulting in improved classification performance [23]. Recent research in GT-based methods has focused primarily on the design of fundamental model frameworks, while also exploring fusion strategies to integrate local and global features, as well as investigating improvements in attention mechanisms [24].

In terms of basic model framework design, numerous research efforts have focused on developing GT models that construct graph structures suitable for Transformer processing. Zhao et al. [25] proposed the GT network with graph attention (GTN-A), which addressed the heterogeneous graph characteristics in HSI by designing methods capable of generating new graph structures, allowing better aggregation of node features. As research progressed, Lu et al. [26] introduced the deformable GT (DGT) with a deformable graph self-attention (DGSA) mechanism that adaptively samples virtual nodes from an HSI constructed from initial graphs. However, preserving detailed spatial and spectral information during graph construction remains a significant challenge [27].

To address the challenge of simultaneously capturing local details and global information when GTs process graph-structured HSI, researchers have proposed various innovative strategies for fusing local and global features. Zheng et al. [22] proposed the GT with spatial–spectral features fusion, which effectively combines local neighborhood features of nodes with global features of long-distance dependencies. Yang et al. [28] developed the GCN and Transformer fusion network (GTFN), which simultaneously explores pixel relationships based on global structural information.

As the core components of GTs, attention mechanisms directly impact the effectiveness of processing graph-structured HSI. Miao et al. [29] proposed a graph-enhanced attention (GA) module that utilizes graph knowledge extracted by the GT module to enhance pixel-level feature representations. Dong et al. [30] introduced the D-LIEG model, which focuses on modeling global spectral–spatial correlations between graph nodes, which improves the discriminability of feature representations. Chen et al. [27] incorporated a contrastive clustering assignment mechanism in the spectral GT (SGT), which maintains cluster consistency by processing nodes in the graph structure and exchanging predicted cluster assignments, enhancing the model's ability to distinguish different spectra. These improvements in attention mechanisms have not only enhanced the capability of GTs to process HSI constructed as graph structures but also improved model robustness to noise and adaptability to limited labeled samples.

B. PE in GT

Self-attention mechanisms are insensitive to input order, causing Transformer models to generate identical vector representations for nodes with different structures but similar features [31]. Transformer models address this problem by incorporating PE to preserve position and sequential relationships of information, achieving significant success in processing sequential data and regular grid data. Early Transformer models employed randomly generated PEs or fixed, nonlearnable PEs, such as sinusoidal PEs (SPEs) [32], [33], while these approaches lack flexibility and struggle to adapt to complex data structures. Methods like absolute PE and rotary

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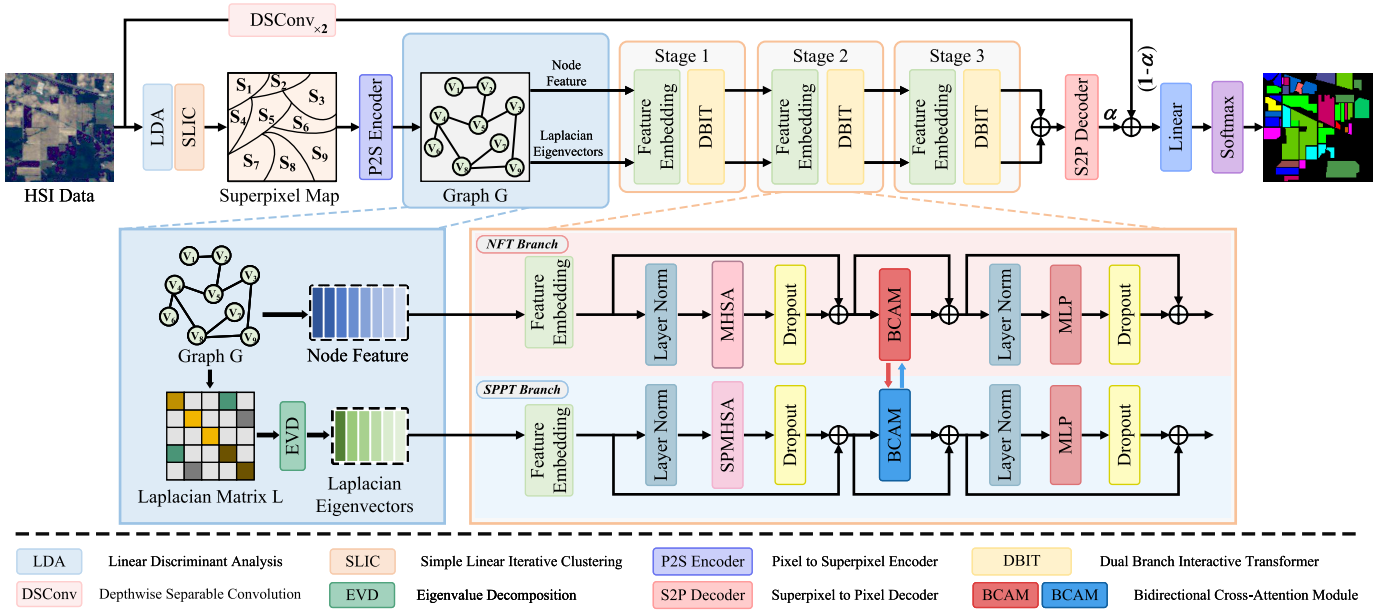


Fig. 1. Overall structure of the SPGFormer framework. SPGFormer primarily consists of three components: the graph construction module, the DBIT module, and the BCAM. The graph construction module segments the original HSI into superpixels and builds a corresponding graph structure. The DBIT module captures rich spatial–spectral information. The BCAM facilitates information exchange between the two branches of the DBIT module, enabling a more comprehensive feature representation.

PE attempt to enhance expressiveness through more sophisticated mathematical formulations. To improve adaptability, researchers proposed learnable PEs, allowing networks to learn position representations best suited for specific tasks during training [34], [35]. The latest development in PE involves conditional approaches, such as the implicit conditional PEs can dynamically generate positional information based on input characteristics [36]. These conditional methods have shown promise in expanding the applicability of Transformers across different domains.

It has been proved that incorporating physical meaning into PEs is particularly important. Following this, center PE [37], spatial PE [38], and spatial–spectral self-attention mechanisms [39] have been developed to create attention mechanisms with both position-aware and spectral-aware capabilities, demonstrating the evolution of PE beyond simple sequential relationships. Furthermore, position self-attention mechanisms assign different weights to different positions to improve discrimination [40]. Although PE has achieved desirable performance in regular grid data, extending these PE techniques to GTs faces unique challenges. In graph-based representation learning frameworks, the positional information of the nodes and topological structures contain crucial information, such as spatial relationships, adjacency relationships, and similarities between the nodes, which are important for HSI classification [41]. Due to the absence of canonical positioning of nodes in arbitrary graph structures, designing PEs for GTs becomes particularly challenging [42].

To address the specific requirements of GTs, researchers have developed various specialized PE methods. Early approaches like absolute PE [32] assign fixed indices to nodes, but this method fails to capture the irregular topology of graph structures. Random walk PE (RWPE) [43] computes positional features based on the landing probabilities of random walks, effectively capturing local neighborhood structures. However, RWPE primarily encodes local connectivity patterns and may overlook global structural properties crucial for HSI classification. Jiang et al. [44] emphasized the importance of PEs, pointing out that node embeddings cannot identify positional information of pixels, while existing methods often randomly generate PEs, ignoring their physical meaning. Researchers have developed various PE strategies to capture both local and global structural properties in GT-based HSI analysis. Among these PE methods, Laplacian PE demonstrates unique advantages [45], [46]. The eigendecomposition of the graph Laplacian matrix produces orthogonal eigenvectors, where each eigenvector corresponds to a specific frequency component of the graph structure, and this spectral decomposition aligns with the inherent structural properties of HSI [47], [48], [49]. Unlike absolute PE that ignores graph topology and RWPE that emphasizes only local patterns, Laplacian PE provides a principled way to encode both local and global structures through its spectral basis. Therefore, Laplacian PE better captures the spatial–spectral structural relationships of HSI while maintaining mathematical rigor, providing more effective positional information representation for GTs.

III. Proposed Method

A. Overall Architecture

The proposed SPGFormer consists of three main modules: the graph construction module, the dual-branch interactive Transformer (DBIT) module, and the BCAM. First, the graph construction module segments the original HSI into superpixels and constructs a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Then, the node feature and the Laplacian eigenvectors are derived from $\mathcal{G}$

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and serve as inputs to the subsequent DBIT module. The DBIT module consists of two parallel processing branches: the NFT branch and the SPPT branch. The NFT branch processes the spectral features through multiple Transformer layers to capture rich spectral dependencies. Simultaneously, the SPPT branch utilizes the graph Laplacian eigenvectors to model global positional information. Furthermore, we propose an SPMHSA mechanism that progressively integrates multiscale k-hop local information as attention bias terms across different network layers. To enhance the interaction between spectral and spatial–structural features, the BCAM serves as a critical connector between the two branches of the DBIT module. Through this bidirectional cross-attention flow, spectral features guide the refinement of PE while structural information enhances spectral feature interpretation, leading to more comprehensive feature representations. Finally, to effectively transition from superpixel-level representations to pixel-level classification, we integrate specialized CNN units based on depthwise separable convolution. These units extract fine-grained features directly from the HSI at the pixel level and intelligently fuse them with the superpixel-level features produced by the DBIT module. The complete architecture of the proposed SPGFormer is illustrated in Fig. 1.

To enhance clarity and readability, Nomenclature summarizes the key notations used throughout this article. The table distinguishes between variables from different branches and processing stages, ensuring a consistent interpretation of the mathematical formulations.

B. Graph Construction and Feature Conversion

1) Superpixels Segmentation: HSI contains not only rich spectral information but also abundant spatial–structural information due to complex land-cover distributions. Modeling these spatial relationships is crucial for effective HSI classification. Representing HSI data as graph structures overcomes the limitations of regular grid representations, allowing flexible modeling of adjacency, similarity, and connectivity patterns between different land covers.

Given a HSI $\mathbf{X}_{\text{in}} \in \mathbb{R}^{h\times w\times c}$, where h, w, and c represent the height, width, and number of spectral bands, respectively, we first perform superpixel segmentation using linear discriminant analysis (LDA) for dimensionality reduction followed by SLIC. The number of superpixels z is determined by the segmentation scale parameter λ

$$z = \frac{h \times w}{\lambda}. \tag{1}$$

This segmentation process transforms the original pixel grid into a set of spatially coherent superpixels $\mathcal{S} = \{\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_z\}$, where each superpixel contains pixels with similar spectral characteristics.

2) Graph Formation: After obtaining the superpixels, we construct an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ represents the set of nodes (superpixels) and $\mathcal{E}$ denotes the edges between adjacent superpixels. For each superpixel, we calculate its node feature by averaging all pixels contained within it

$$\mathbf{F}_i = \frac{1}{n_i}\sum_{j=1}^{n_i} \mathbf{X}_i^j \tag{2}$$

where $\mathbf{F}_i$ represents the feature vector of the ith superpixel node, $\mathbf{X}_i^j$ denotes the jth pixel in the ith superpixel, and n_i is the total number of pixels in ith superpixel. The edge weights in our graph reflect the similarity between adjacent superpixels and are calculated using a Gaussian kernel function

$$\mathbf{W}_{ij} = \begin{cases} \exp\left(-\frac{\|\mathbf{F}_i - \mathbf{F}_j\|_2^2}{\sigma^2}\right), & \text{if } \mathcal{S}_i \text{ and } \mathcal{S}_j \text{ are adjacent} \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where σ is a parameter controlling the decay rate and $\|\mathbf{F}_i - \mathbf{F}_j\|_2$ represents the Euclidean distance between superpixels nodes.

3) Pixel–Superpixel Conversion Decoders: Since our model operates on superpixel-level representations while the classification of HSI requires pixel-level predictions, we need a mechanism to convert between pixel and superpixel. To address this, we define an association matrix $\mathbf{M} \in \mathbb{R}^{hw\times z}$ that establishes the membership relationships between pixels and superpixels

$$\mathbf{M}_{i,j} = \begin{cases} 1, & \text{if } i\text{th pixel belongs to } j\text{th superpixel} \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

This binary matrix $\mathbf{M}$ serves as the foundation for our bidirectional conversion between pixel and superpixel representations.

Based on the association matrix, we implement two dedicated conversion modules, pixel-to-superpixel (P2S) decoder and superpixel-to-pixel (S2P) decoder. The P2S decoder is applied after superpixel segmentation for graph construction, enabling efficient graph-based processing while preserving essential structural information. The S2P decoder is used at the final stage of our model to convert the learned superpixel-level features back to pixel-level representations, facilitating pixelwise classification required for HSI analysis. The feature of superpixels can be represented as

$$\mathbf{F} = \text{P2S}(\mathbf{X}, \mathbf{M}) = \hat{\mathbf{M}}^{\text{T}} \cdot \text{Flatten}(\mathbf{X}) \tag{5}$$

where $\hat{\mathbf{M}}$ is the column-normalized $\mathbf{M}$ and Flatten($\mathbf{X}$) reshapes the spatial dimensions of the input tensor $\mathbf{X}$ into a single dimension. This normalization ensures that superpixels of different sizes receive balanced feature representations.

Correspondingly, the S2P decoder transforms superpixel-level features back to the pixel domain

$$\mathbf{X} = \text{S2P}(\mathbf{F}, \mathbf{M}) = \text{Reshape}(\mathbf{M} \cdot \mathbf{F}) \tag{6}$$

where Reshape($\cdot$) restores the spatial dimensions of the flattened data.

C. SPPT Branch

As illustrated in Fig. 1 (SPPT Branch), the proposed SPPT branch effectively integrates global graph position information with local topological structure information, achieving hierarchical spatial relationship modeling. This module consists of three key components: Laplacian PE, k-hop local structure embedding, and SPMHSA.

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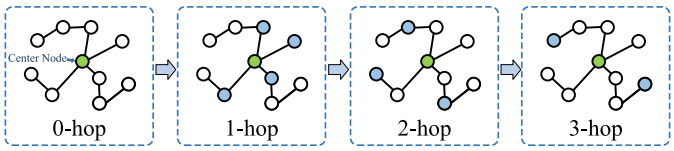


Fig. 2. Schematic of the hop graph structure. The green node indicates the center node, and the blue nodes are its k-hop neighbors.

1) Laplacian PE: Based on the weighted adjacency matrix $\mathbf{W}$, we build the graph Laplacian matrix $\mathbf{L}$ and calculate its eigenvectors as PE. Specifically, we first construct the degree matrix $\mathbf{D}$, where $D_{ii} = \sum_j W_{ij}$. The normalized Laplacian matrix is defined as follows:

$$\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2} \tag{7}$$

where $\mathbf{I}$ is the identity matrix. We then perform eigendecomposition on the Laplacian matrix $\mathbf{L}$

$$\mathbf{L} = \mathbf{U}\Lambda\mathbf{U}^{\mathrm{T}} \tag{8}$$

where $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_z)$ is a diagonal matrix of eigenvalues (sorted from smallest to largest), z is the number of superpixels and $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_z]$ represents the corresponding eigenvectors. While eigendecomposition typically has $\mathcal{O}(z^3)$ computational complexity, the Laplacian matrix $\mathbf{L}$ exhibits sparsity due to the local connectivity of the superpixel graph. More importantly, since only the first m smallest eigenvectors are required for PE, efficient sparse partial eigendecomposition can be employed to significantly reduce the computational burden. From a spectral graph theory perspective, the graph Laplacian eigenvectors provide a natural basis for functions defined on graphs, where lower frequency eigenvectors (corresponding to smaller eigenvalues) capture global smooth variations while higher frequency ones encode local structural details. This property is particularly well-suited for HSI, where spatially adjacent pixels often belong to the same land-cover class while boundaries between different materials create local discontinuities. Unlike absolute PE that assumes fixed grid coordinates or random PE that lacks structural meaning, Laplacian PE inherently encodes the connectivity patterns and similarity relationships in the HSI graph structure, making it particularly suitable for preserving spatial–spectral topology.

It should be noted that the first eigenvalue of the Laplacian matrix $\lambda_1 = 0$, and its corresponding eigenvector $\mathbf{u}_1$ is a constant vector containing no useful structural information. Therefore, we choose to start from the second eigenvector, taking m eigenvectors to form the PE matrix

$$\mathbf{P} = \left[\mathbf{u}_2, \mathbf{u}_3, \ldots, \mathbf{u}_{m+1}\right] \in \mathbb{R}^{z\times m}. \tag{9}$$

Unlike traditional PEs predefined based on coordinate relationships in Euclidean space for sequential data or regular grid data, Laplacian eigenvector-based positional information derives from the topological structure of the graph itself, naturally incorporating intrinsic spatial association patterns and global structural information of HSI.

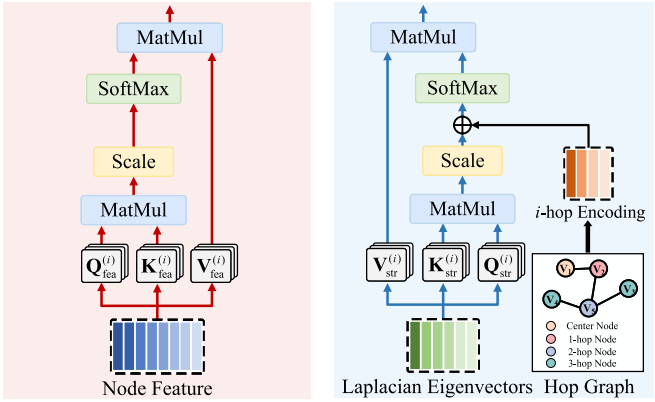


Fig. 3. Structure diagram of (a) MHSA mechanism and (b) SPMHSA mechanism.

2) k-Hop Local Structure: The distribution of land cover in HSI exhibits distinct local and global characteristics. Therefore, modeling both local and global features of nodes is crucial for HSI classification. In this article, we employ k-hop to model multiscale local topological structures of nodes. The local topological structure of the k hop is illustrated in Fig. 2. Specifically, given the adjacency matrix $\mathbf{W}$, we construct k-hop adjacency matrices through depth-first search (DFS). For each node v, DFS finds all paths that reach other nodes in exactly k steps. The k-hop adjacency matrix is defined as

$$\mathbf{A}_k(v, v_k) = \frac{1}{k}\left(\mathbf{W}(v, v_1) + \mathbf{W}(v_1, v_2) + \cdots + \mathbf{W}(v_{k-1}, v_k)\right) \tag{10}$$

where $v_1, v_2, \ldots, v_{k-1}$ are the intermediate nodes from v to v_k along a k-hop path. Specifically, these nodes represent the sequence of superpixels traversed in order during the DFS traversal. v_1 is the first node adjacent to v on the path, v_2 is the second node adjacent to v_1, and v_{k-1} being the $(k-1)$th node that is adjacent to the final destination. When there are multiple paths between v and v_k, we select the path with the highest average edge weight. Based on this method, we construct k-hop graph adjacency matrices $\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_k$. This k-hop graph structure expands the receptive field of a single node, enabling it to establish connections with more nodes through adjacent nodes. We use these k-hop ($k = 3$) adjacency matrices as structure information matrices

$$\mathbf{S}^{(1)} = \mathbf{A}_1,\ \mathbf{S}^{(2)} = \mathbf{A}_2,\ \mathbf{S}^{(3)} = \mathbf{A}_3. \tag{11}$$

To effectively model multiscale topological structures, we progressively incorporate structural information with increasing hop numbers (from 1 to 3) as the network depth increases. This hierarchical local structure information fusion strategy enables the model to gradually expand its receptive field with network depth, thereby capturing the spatial structural relationships in HSI more comprehensively.

3) Structure Perception Multihead Self-Attention: As illustrated in Fig. 3(b), the proposed SPMHSA combines k-hop local structure information with the standard self-attention mechanism, enhancing the model's structure-aware capability. Given the input features of SPPT branch $\mathbf{H}_{\mathrm{str}}^{(i)} \in \mathbb{R}^{z\times d}$ ith layer,

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we first generate Query($\mathbf{Q}$), Key($\mathbf{K}$), and Value($\mathbf{V}$) matrices through linear transformations

$$\begin{aligned}\mathbf{Q}_{\text{str}}^{(i)} &= \text{LN}\left(\mathbf{H}_{\text{str}}^{(i)}\mathbf{W}_{Q,\text{str}}^{(i)}\right)\\ \mathbf{K}_{\text{str}}^{(i)} &= \text{LN}\left(\mathbf{H}_{\text{str}}^{(i)}\mathbf{W}_{K,\text{str}}^{(i)}\right)\\ \mathbf{V}_{\text{str}}^{(i)} &= \text{LN}\left(\mathbf{H}_{\text{str}}^{(i)}\mathbf{W}_{V,\text{str}}^{(i)}\right)\end{aligned} \tag{12}$$

where $\mathbf{W}_{Q,\text{str}}^{(i)}$, $\mathbf{W}_{K,\text{str}}^{(i)}$, and $\mathbf{W}_{V,\text{str}}^{(i)}$ are learnable parameter matrices, and LN represents the layer normalization operation. Unlike traditional self-attention, we incorporate the structure information $\mathbf{S}^{(i)}$ into the calculation of attention scores, and the SPMHSA can be expressed as

$$\begin{aligned}\mathbf{A}_{\text{str}}^{(i)} &= \text{SPMHSA}\left(\mathbf{Q}_{\text{str}}^{(i)}, \mathbf{K}_{\text{str}}^{(i)}, \mathbf{V}_{\text{str}}^{(i)}, \mathbf{S}^{(i)}\right)\\ &= \text{Softmax}\left(\frac{\mathbf{Q}_{\text{str}}^{(i)}\left(\mathbf{K}_{\text{str}}^{(i)}\right)^{\text{T}}}{\sqrt{d_k}} + \mathbf{S}^{(i)}\right)\mathbf{V}_{\text{str}}^{(i)}\end{aligned} \tag{13}$$

where SPMHSA$(\cdot)$ presents the SPMHSA, d_k is the dimension of key vector $\mathbf{K}_{\text{str}}^{(i)}$. Unlike traditional multiscale feature fusion methods that extract features at different scales separately and then fuse them through postprocessing strategies, SPMHSA directly integrates multiscale structural information into the attention computation process. Specifically, SPMHSA incorporates k-hop structural information as attention bias terms, enabling the model to progressively capture spatial relationships from local to global scales across different network layers. This structure-aware attention mechanism achieves multiscale perception within a unified framework, offering both mathematical elegance and computational efficiency for comprehensively understanding the complex spatial structures in HSI. Finally, the dropout regularization and residual connection are employed, and the output of the SPPT branch can be represented as

$$\mathbf{O}_{\text{str}}^{(i)} = \mathbf{H}_{\text{str}}^{(i)} + \text{Dropout}\left(\mathbf{A}_{\text{str}}^{(i)}\right). \tag{14}$$

D. NFT Branch

As illustrated in Fig. 1, the node feature branch adopts a standard multihead self-attention Transformer framework, specifically including linear projection layers, multihead self-attention mechanism, feed-forward unit, and residual connections, used for processing spectral feature information of superpixel nodes. For the input superpixel node feature matrix $\mathbf{H}_{\text{fea}}^{(i)}$

$$\begin{aligned}\mathbf{Q}_{\text{fea}}^{(i)} &= \text{LN}\left(\mathbf{H}_{\text{fea}}^{(i)}\mathbf{W}_{Q,\text{fea}}^{(i)}\right)\\ \mathbf{K}_{\text{fea}}^{(i)} &= \text{LN}\left(\mathbf{H}_{\text{fea}}^{(i)}\mathbf{W}_{K,\text{fea}}^{(i)}\right)\\ \mathbf{V}_{\text{fea}}^{(i)} &= \text{LN}\left(\mathbf{H}_{\text{fea}}^{(i)}\mathbf{W}_{V,\text{fea}}^{(i)}\right).\end{aligned} \tag{15}$$

The standard multihead self-attention mechanism can be expressed as

$$\mathbf{A}_{\text{fea}}^{(i)} = \text{MHSA}\left(\mathbf{Q}_{\text{fea}}^{(i)}, \mathbf{K}_{\text{fea}}^{(i)}, \mathbf{V}_{\text{fea}}^{(i)}\right)$$

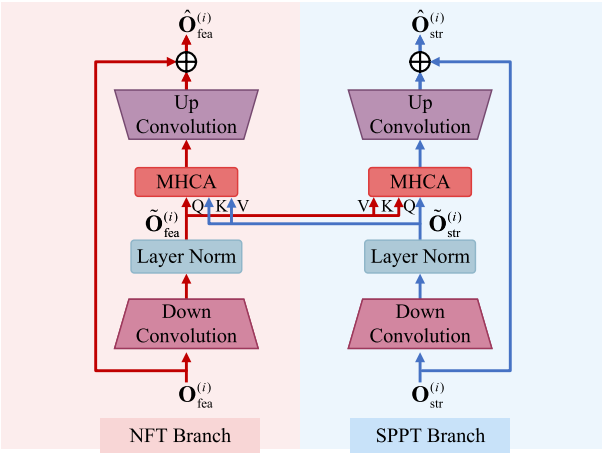


Fig. 4. Structure diagram of BCAM. The MHCA is employed to enable information interaction between the two branches.

$$= \text{Softmax}\left(\frac{\mathbf{Q}_{\text{fea}}^{(i)}\left(\mathbf{K}_{\text{fea}}^{(i)}\right)^{\text{T}}}{\sqrt{d_k}}\right)\mathbf{V}_{\text{fea}}^{(i)}. \tag{16}$$

Similarly, through dropout regularization and residual connection, the output of the node feature self-attention can be represented as

$$\mathbf{O}_{\text{fea}}^{(i)} = \mathbf{H}_{\text{fea}}^{(i)} + \text{Dropout}\left(\mathbf{A}_{\text{fea}}^{(i)}\right). \tag{17}$$

E. Bidirectional Cross-Attention Module

To achieve effective information exchange between the NFT branch and the SPPT branch, we propose a BCAM, as illustrated in Fig. 4. Unlike classical self-attention, BCAM allows queries to come from one branch, while the keys and values come from another branch. We first define the multihead cross-attention (MHCA) mechanism as follows:

$$\begin{aligned}&\text{MHCA}\left(\mathbf{X}_1, \mathbf{X}_2\right)\\ &= \text{Softmax}\left(\frac{\left(\mathbf{X}_1\mathbf{W}_{Q,\text{cro}}\right)\left(\mathbf{X}_2\mathbf{W}_{K,\text{cro}}\right)^{\text{T}}}{\sqrt{d_k}}\right)\left(\mathbf{X}_2\mathbf{W}_{V,\text{cro}}\right)\end{aligned} \tag{18}$$

where $\mathbf{X}_1$ and $\mathbf{X}_2$ represent features from different branches, with $\mathbf{X}_1$ providing queries and $\mathbf{X}_2$ providing keys and values. $\mathbf{W}_{Q,\text{cro}}$, $\mathbf{W}_{K,\text{cro}}$, and $\mathbf{W}_{V,\text{cro}}$ are learnable projection matrices. However, directly applying the cross-attention mechanism may lead to excessive computational overhead, especially when processing high-dimensional features. To address this challenge, we introduce 1×1 convolutions to achieve efficient feature dimension transformation, proposing a convolutional bottleneck cross-attention design. Specifically, we first downsample the input features using 1×1 convolution, then normalize the data through LN layers, followed by MHCA between the two branches, and finally upsample using 1×1 convolution. In addition, residual connections are used to preserve the original features of both branches.

Based on the BCAM definition and the efficiency considerations discussed above, the specific implementation for the SPPT branch can be formulated as follows:

$$\hat{\mathbf{O}}_{\text{str}}^{(i)} = \mathbf{O}_{\text{str}}^{(i)} + \text{ConvU}\left(\text{MHCA}\left(\tilde{\mathbf{O}}_{\text{str}}^{(i)}, \tilde{\mathbf{O}}_{\text{fea}}^{(i)}\right)\right). \tag{19}$$

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Similarly, the BCAM of the NFT branch can be represented as

$$\hat{\mathbf{O}}_{\text{fea}}^{(i)} = \mathbf{O}_{\text{fea}}^{(i)} + \text{ConvU}\left(\text{MHCA}\left(\tilde{\mathbf{O}}_{\text{fea}}^{(i)}, \tilde{\mathbf{O}}_{\text{str}}^{(i)}\right)\right) \tag{20}$$

where $\tilde{\mathbf{O}}_{\text{fea}}^{(i)} = \text{LN}(\text{ConvD}(\mathbf{O}_{\text{fea}}^{(i)}))$, $\tilde{\mathbf{O}}_{\text{str}}^{(i)} = \text{LN}(\text{ConvD}(\mathbf{O}_{\text{str}}^{(i)}))$, ConvD and ConvU are downsampling and upsampling 1×1 convolution layers, LN represents the LayerNorm layer. The proposed BCA makes the SPPT branch fully utilize key semantic information captured by the NFT branch to guide its structural understanding, enhancing the ability to distinguish regions with similar structures but different semantics. Correspondingly, the NFT branch can focus on spatial topological information provided by the SPPT branch, enhancing the understanding of global and local structural information.

F. Feature Fusion and Classification

Following the BCAM, the outputs of these two branches undergo LN, MLP, and dropout layers to form the output of the $(i+1)$th layer

$$\mathbf{H}_{\text{str}}^{(i+1)} = \hat{\mathbf{O}}_{\text{str}}^{(i)} + \text{Dropout}\left(\text{LN}\left(\text{MLP}\left(\hat{\mathbf{O}}_{\text{str}}^{(i)}\right)\right)\right) \tag{21}$$

$$\mathbf{H}_{\text{fea}}^{(i+1)} = \hat{\mathbf{O}}_{\text{fea}}^{(i)} + \text{Dropout}\left(\text{LN}\left(\text{MLP}\left(\hat{\mathbf{O}}_{\text{fea}}^{(i)}\right)\right)\right). \tag{22}$$

To fully leverage the complementary information captured by the dual-branch architecture, we fuse the features derived from these two branches at the last layer

$$\mathbf{H}_{\text{trs}} = \mathbf{H}_{\text{str}} + \mathbf{H}_{\text{fea}}. \tag{23}$$

It is worth noting that relying solely on these two superpixels-level Transformer branches may struggle to capture pixel-level detail features. To address this, we introduce a dedicated CNN branch that extracts pixel-level features through two layers of depthwise separable convolution

$$\mathbf{H}_{\text{cnn}} = \text{DSConv}_2\left(\text{DSConv}_1\left(\mathbf{X}_{\text{in}}\right)\right) \tag{24}$$

where DSConv$(\cdot)$ represents depthwise separable convolution and $\mathbf{X}_{\text{in}}$ is the input HSI. To adaptively fuse these two features, we introduce a learnable weight parameter α for weighted fusion

$$\mathbf{H}_{\text{fus}} = \alpha \cdot \mathbf{H}_{\text{trs}} + (1-\alpha) \cdot \mathbf{H}_{\text{cnn}} \tag{25}$$

where α is a learnable scalar parameter, initialized to 0.5, and gradually optimized to the optimal value during the training process.

Finally, the fused features $\mathbf{H}_{\text{fus}}$ are fed into the classification layer and use the cross-entropy loss function for model optimization

$$\mathcal{L}_{\text{CE}} = -\sum_{i=1}^{c} y_i \log\left(\hat{y}_i\right) \tag{26}$$

where y_i is the one-hot encoding of the true label, $\hat{y}_i$ is the category probability predicted by the model, and c is the total number of categories.

IV. Experimental Results

A. Datasets Description

To comprehensively evaluate the performance of our proposed SPGFormer method, we conducted experiments on four widely used and diverse HSI datasets: University of Pavia (UP), Salinas (SA), WHU-Hi-LongKou (LK), and Houston University 2013 (HU). These datasets represent a variety of remote sensing scenarios with different spatial resolutions, spectral characteristics, and land-cover types. Detailed information about the dataset are shown in Table I.

1) University of Pavia: This dataset was captured by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over Pavia University in northern Italy. It consists of 610×340 pixels with a spatial resolution of 1.3 m/pixel and contains 103 spectral bands after removing bands with low signal-to-noise ratio.

2) Salinas: The SA dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the Salinas Valley, CA, USA, with a spatial resolution of 3.7 m. The image comprises 512×217 pixels and 204 spectral bands after removing water absorption bands.

3) WHU-Hi-LongKou: This dataset was acquired using an autonomous aerial vehicle (AAV) equipped with a hyperspectral sensor over an agricultural zone in Hubei, China. The image contains 550×400 pixels with 270 spectral bands ranging from 0.4 to 1.0 μm. The ground truth covers nine land-cover classes, primarily consisting of agricultural areas and natural vegetation.

4) Houston University 2013: The HU dataset was collected by the Compact Airborne Spectrographic Imager (CASI-1500) over the University of Houston campus and its surrounding areas in Houston, TX, USA, with a spatial resolution of 2.5 m. The image has 349×1905 pixels and 144 spectral bands ranging from 0.38 to 1.05 μm. The ground truth includes 15 urban and suburban land-cover categories.

B. Experimental Setup

1) Implementation Details: The comparative experiments and SPGFormer utilize the Adam optimizer for training, with the learning rate set to 5×10^{-4} and the epoch set to 200. The specific network structure is consistent with the design outlined in the original paper. The code was executed in Python-3.11 and PyTorch 2.2.2+cuda 12.1, and the experimental environment consists of an i5-12490F CPU and an RTX-3060Ti GPU. For each dataset, 15 samples per class were randomly selected for the training set, and the remaining samples were designated for the testing set.

2) Evaluation Metrics: In this experiment, four evaluation metrics that are widely used for HSI classification were employed. These metrics include the overall accuracy (OA), average accuracy (AA), category accuracy (CA), and the Kappa coefficient (Kappa). To eliminate the impact of randomness, all experiments were performed ten repeated tests, and the mean and standard deviation values of each evaluation metric were reported. In addition, we provide full classification maps with critical region zoom-ins. These maps display predicted labels for every pixel in the HSI and enable

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TABLE I
DETAILS OF EACH LAND-COVER CLASS OF FOUR DATASETS USED FOR EVALUATION IN THE EXPERIMENTS

No.	University of Pavia			Salinas			WHU-Hi-LongKou			Houston University		
	Class Name	Training	Testing	Class Name	Training	Testing	Class Name	Training	Testing	Class Name	Training	Testing
1	Asphalt	15	6616	Brocoli_green_weeds_1	15	1994	Corn	15	34496	Healthy_grass	15	1359
2	Meadows	15	18634	Brocoli_green_weeds_2	15	3711	Cotton	15	8359	Stressed_grass	15	1439
3	Gravel	15	2084	Fallow	15	1961	Sesame	15	3016	Synthetic_grass	15	745
4	Trees	15	3049	Fallow_rough_plow	15	1379	Broad-leaf soybean	15	63197	Trees	15	1249
5	Metal-sheets	15	1330	Fallow_smooth	15	2663	Narrow-leaf soybean	15	4136	Soil	15	1283
6	Bare-soil	15	5014	Stubble	15	3944	Rice	15	11839	Water	15	310
7	Bitumen	15	1315	Celery	15	3564	Water	15	67041	Residential	15	1461
8	Bricks	15	3667	Grapes_untrained	15	11256	Roads and houses	15	7109	Commercial	15	1339
9	Shadows	15	932	Soil_vinyard_develop	15	6188	Mixed weed	15	5214	Road	15	1539
10				Corn_senesced_green_weeds	15	3263				Highway	15	1409
11				Lettuce_romaine_4wk	15	1053				Railway	15	1498
12				Lettuce_romaine_5wk	15	1912				Parking_lot_1	15	1414
13				Lettuce_romaine_6wk	15	901				Parking_lot_2	15	555
14				Lettuce_romaine_7wk	15	1055				Tennis_court	15	466
15				Vinyard_untrained	15	7253				Running_track	15	743
16				Vinyard_vertical_trellis	15	1792						

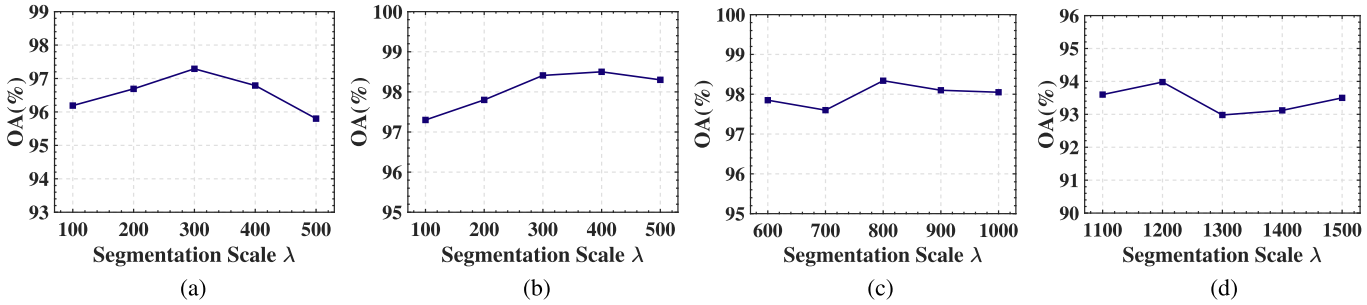


Fig. 5. Classification accuracies of SPGFormer with different segmentation scales on each dataset. (a) UP. (b) SA. (c) LK. (d) HU.

intuitive visual comparison of different methods, particularly regarding spatial consistency and boundary preservation [50].

3) Baseline: To evaluate the performance of SPGFormer, we compare it with the following state-of-the-art (SOTA) methods: the attention multihop graph and multiscale convolutional fusion network (AMGCFN) [51], the groupwise separable convolutional vision transformer (GSC-ViT) network [52], the memory-augmented spectral–spatial transformer (MASSFormer) network [53], the S^2GFormer framework [3], the PSFormer network [4], the double branch convolution-transformer (DBCT) network [54], the graph-in-graph convolutional network (GiGCN) [55], the non-Euclidean spectral–spatial feature mining network with gated GCN–CNN (NESSGGCN) [56], and the spatial–spectral enhancement and fusion network (SSEFN) [57].

C. Hyperparameter Analysis

1) Influence of Superpixel Scale: Fig. 5 illustrates the OAs of our SPGFormer under different superpixel segmentation scales (λ) across four datasets. It's important to clarify that larger λ values result in fewer superpixels, with each superpixel containing more pixels, and vice versa. It can be observed from Fig. 5 that for smaller datasets with high intraclass consistency, such as SA (512×217), a relatively small superpixel scale ($\lambda = 400$) can achieve optimal performance. For the UP dataset (610×340) with more complex spatial structures, despite its moderate size, a smaller superpixel scale ($\lambda = 300$) is required to preserve detailed information due to the complex boundaries and small objects in this urban scene. For larger datasets like LK (550×400) and HU (349×1905), we need larger superpixel scales. The LK dataset, primarily an agricultural region with strong intraclass consistency, is suitable for medium-sized superpixels ($\lambda = 800$), while the HU dataset, containing complex urban structures and diverse land-cover types, requires a larger superpixel scale ($\lambda = 1200$) to balance local details and overall structure. Overall, the selection of superpixel scale needs to balance two factors: the spatial size of HSI and its internal complexity. For HSIs of similar size, those with better data distribution consistency can typically use larger λ, as similar regions can be effectively grouped into the same superpixel; while highly heterogeneous scenes with complex boundaries require smaller λ values to preserve boundary information. This finding provides practical guidance for selecting superpixel parameters in different scenarios.

2) Influence of Net-Depth and Self-Attention Heads: Fig. 6 provides a detailed illustration of SPGFormer's classification accuracy on four datasets under varying numbers of transformer encoder layers and multihead self-attention heads. The results demonstrate that the proposed SPGFormer exhibits strong robustness to hyperparameter variations, maintaining high performance across different configurations. Specifically, on the UP dataset, the three-layer and eight-head configuration achieves an OA of 97.29%, approximately 1.2% higher than the two-layer and eight-head setup; on the SA dataset, the same configuration reaches an OA of 98.41%; similar trends are observed for the LK and HU datasets. This consistency indicates that this configuration can effectively adapt to HSI with different characteristics, showing excellent

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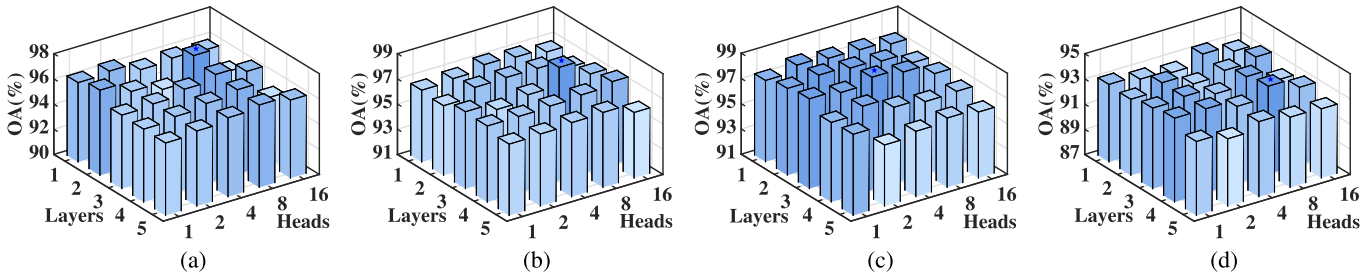


Fig. 6. Classification accuracies of SPGFormer with different numbers of transformer encoder layers and multihead self-attention heads on each dataset. (a) UP. (b) SA. (c) LK. (d) HU.

TABLE II

QUANTITATIVE RESULT (ACC% ± STD%) OF UP. BEST IN BOLD AND SECOND WITH UNDERLINE

Class	AMGCFN TGRS2023	GSC-ViT TGRS2024	MASSFormer TGRS2024	S^{2}GFormer TGRS2024	PSFormer TGRS2024	DBCT TGRS2024	GiGCN TNNLS2024	NESSGGCN ESWA2025	SSEFN TGRS2025	SPGFormer Ours
1	95.47±2.69	77.30±15.79	84.05±5.86	70.96±1.69	96.67±1.89	82.79±9.64	**99.31±0.71**	90.30±4.25	97.68±1.77	93.54±2.17
2	89.38±3.72	85.80±14.75	91.70±3.54	90.12±1.55	87.27±5.27	**98.75±1.20**	77.90±8.24	97.67±1.54	91.42±9.64	97.56±2.85
3	97.40±1.00	72.70±21.75	95.11±2.22	84.56±0.13	94.43±4.78	97.50±1.37	**100.00±0.00**	84.21±4.32	**100.00±0.00**	97.55±1.01
4	95.76±6.42	95.08±5.52	95.87±1.54	93.93±0.91	96.06±0.76	**96.85±0.89**	96.52±1.42	93.67±2.16	94.76±1.97	94.94±1.37
5	**100.00±0.00**	**100.00±0.00**	97.74±0.57	99.65±0.25	**100.00±0.00**	99.47±0.42	99.92±0.08	99.25±0.25	99.97±0.03	99.24±0.32
6	**100.00±0.00**	83.35±5.55	57.62±16.23	89.50±1.48	99.27±1.15	91.26±4.32	99.94±0.06	91.36±3.69	99.98±0.02	99.90±0.26
7	99.69±0.09	99.70±0.53	99.01±0.84	95.44±1.74	99.89±0.07	92.53±5.64	**100.00±0.00**	96.58±2.58	**99.97±0.13**	**100.00±0.00**
8	97.59±3.74	98.23±1.98	93.02±5.21	63.85±8.85	**99.42±0.30**	65.39±9.82	94.13±2.54	85.03±7.11	97.07±2.13	98.58±3.69
9	99.59±0.75	99.36±1.02	93.56±4.59	98.25±0.36	99.63±0.46	99.57±0.29	99.46±0.16	94.74±2.14	99.49±0.05	**100.00±0.00**
OA	94.01±1.26	86.45±4.85	87.54±4.27	85.45±3.57	93.22±2.46	92.18±4.67	89.46±3.12	93.70±2.54	95.25±1.12	**97.29±1.18**
AA	97.24±0.72	90.17±3.36	89.74±2.65	87.36±1.16	96.96±1.06	91.57±4.33	96.35±2.14	92.53±3.68	97.81±1.05	**97.92±1.01**
Kappa	92.22±1.62	82.44±6.81	83.48±7.55	81.08±1.70	91.26±3.11	89.64±4.28	86.63±5.44	91.64±2.18	92.19±5.18	**96.43±2.54**

TABLE III

QUANTITATIVE RESULT (ACC% ± STD%) OF SA. BEST IN BOLD AND SECOND WITH UNDERLINE

Class	AMGCFN TGRS2023	GSC-ViT TGRS2024	MASSFormer TGRS2024	S^{2}GFormer TGRS2024	PSFormer TGRS2024	DBCT TGRS2024	GiGCN TNNLS2024	NESSGGCN ESWA2025	SSEFN TGRS2025	SPGFormer Ours
1	99.69±0.09	88.32±13.35	**100.00±0.00**	99.82±0.13	**100.00±0.00**	**100.00±0.00**	99.73±0.24	99.95±0.05	**100.00±0.00**	**100.00±0.00**
2	**100.00±0.00**	91.09±12.88	66.29±8.44	**100.00±0.00**	99.91±0.17	98.68±2.18	**100.00±0.00**	**100.00±0.00**	99.91±0.09	**100.00±0.00**
3	**100.00±0.00**	96.38±4.58	**100.00±0.00**	99.97±0.05	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	98.47±2.14	**100.00±0.00**	**100.00±0.00**
4	99.20±0.33	99.30±1.04	**100.00±0.00**	98.91±0.31	99.95±0.03	99.20±0.53	**100.00±0.00**	99.93±0.07	99.92±0.08	99.92±0.08
5	92.21±1.85	98.30±1.16	99.40±0.42	94.54±0.65	98.53±0.97	99.77±0.43	**99.92±0.08**	94.44±4.56	98.49±0.57	99.28±0.74
6	99.08±0.37	99.92±0.12	**100.00±0.00**	98.22±0.37	99.89±0.09	99.95±0.05	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
7	**100.00±0.00**	99.46±0.72	99.49±1.05	98.54±0.28	99.94±0.02	**100.00±0.00**	99.53±0.38	97.25±1.15	**100.00±0.00**	**100.00±0.00**
8	87.59±3.52	76.77±10.99	79.97±7.25	86.45±1.31	86.87±4.29	82.72±8.18	77.11±14.19	89.85±5.64	89.54±5.17	**97.14±1.62**
9	**100.00±0.00**	85.47±7.97	99.66±1.12	**100.00±0.00**	99.90±0.19	99.08±0.54	**100.00±0.00**	99.90±0.08	**100.00±0.00**	**100.00±0.00**
10	91.54±3.47	95.21±2.18	93.96±4.13	97.11±0.35	96.02±2.63	92.74±5.12	94.27±4.62	**98.10±3.17**	96.29±3.66	96.41±2.14
11	99.81±0.30	93.73±10.83	99.81±0.32	**100.00±0.00**	99.03±1.22	99.15±0.81	**100.00±0.00**	98.20±1.26	**100.00±0.00**	**100.00±0.00**
12	**100.00±0.00**	99.42±0.81	99.95±0.05	99.83±0.07	**100.00±0.00**	99.69±0.53	99.87±0.09	**100.00±0.00**	99.21±0.49	**100.00±0.00**
13	99.66±0.14	**100.00±0.00**	**100.00±0.00**	99.85±0.14	99.33±0.29	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
14	99.41±0.57	99.56±0.24	99.43±0.46	98.26±1.19	99.05±1.02	99.62±0.45	**100.00±0.00**	**100.00±0.00**	98.29±1.25	99.71±0.34
15	92.01±3.49	83.56±16.49	94.33±3.58	87.44±1.34	92.70±2.50	91.64±7.52	**98.99±1.84**	83.06±7.67	89.44±5.36	94.70±3.36
16	**100.00±0.00**	97.86±1.80	99.67±0.47	97.81±1.11	98.93±0.42	96.04±2.42	99.42±0.35	99.16±1.31	99.88±0.07	99.49±0.51
OA	95.32±1.11	89.43±4.71	92.24±3.61	94.66±0.23	95.85±0.72	94.43±3.65	94.73±2.41	94.90±3.65	96.02±1.58	**98.41±1.01**
AA	97.55±0.62	94.02±3.67	95.75±2.69	97.30±0.07	98.13±0.20	97.39±1.07	98.14±1.31	97.39±1.21	98.13±1.24	**99.16±0.42**
Kappa	94.80±1.23	88.28±5.22	91.38±4.53	94.05±0.26	95.38±0.80	93.81±4.11	94.16±2.39	94.32±3.34	95.57±2.69	**98.23±1.55**

generalizability. Overall, the model generally achieves optimal results with three transformer layers and eight attention heads for all datasets. Although there are slight performance variations between datasets, this configuration consistently provides the best balance between model expressiveness and computational efficiency. To simplify the experimental design and maintain consistency, we adopt this configuration (three layers and eight heads) for all subsequent experiments. This standardized setting also demonstrates the practical applicability of our method without requiring extensive dataset-specific hyperparameter tuning.

D. Classification Results

1) Comparative Experimental Analysis: Tables II–V evaluate the classification performance of our proposed SPGFormer against nine SOTA baselines on the UP, SA, LK, and HU datasets. Overall, the proposed SPGFormer method achieves the best results across all four datasets in terms of OA, AA, and the Kappa coefficient. Among GCN-based methods, GiGCN shows strong node information aggregation but weaker pixel-level feature extraction, achieving OA values of 89.46%, 94.73%, 94.60%, and 86.53% on UP, SA, LK, and

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TABLE IV
QUANTITATIVE RESULT (ACC% ± STD%) OF LK. BEST IN BOLD AND SECOND WITH UNDERLINE

Class	AMGCFN TGRS2023	GSC-ViT TGRS2024	MASSFormer TGRS2024	S^2GFormer TGRS2024	PSFormer TGRS2024	DBCT TGRS2024	GiGCN TNNLS2024	NESSGGCN ESWA2025	SSEFN TGRS2025	SPGFormer Ours
1	96.62±3.19	98.94±1.10	94.33±2.73	99.85±0.18	98.29±1.02	98.91±1.12	96.11±2.18	**99.94±0.06**	98.43±1.12	99.28±0.75
2	**100.00±0.00**	80.36±7.41	99.04±0.71	99.98±0.02	99.78±0.18	96.29±2.17	99.19±0.67	99.87±0.24	90.39±5.43	97.43±2.44
3	96.61±2.49	84.68±4.83	96.31±2.45	98.64±1.15	99.93±0.07	99.46±0.31	96.51±3.19	**100.00±0.00**	74.63±11.89	**100.00±0.00**
4	89.95±7.27	92.89±3.18	95.15±3.42	92.42±4.14	94.78±1.56	95.02±2.46	89.81±5.23	93.76±5.77	**98.95±1.71**	96.25±2.89
5	99.63±0.33	95.72±3.14	99.63±0.18	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.85±0.16	99.18±0.79	81.26±9.34	99.68±0.54
6	99.78±0.18	92.65±3.46	**99.89±0.11**	99.88±0.07	99.82±0.18	99.86±0.11	99.64±0.33	95.56±3.18	99.68±0.28	99.77±0.24
7	98.64±1.12	99.46±0.78	99.71±0.24	99.82±0.13	94.23±3.41	99.79±0.21	99.75±0.20	99.90±0.08	98.97±0.88	**99.96±0.04**
8	84.35±10.53	95.01±1.54	88.42±4.75	91.36±3.73	83.96±8.15	90.54±4.12	82.74±5.26	**97.36±1.65**	89.26±4.27	95.03±2.17
9	96.49±1.72	56.98±13.83	69.67±10.57	83.85±5.53	96.16±2.11	81.68±6.34	68.71±12.66	77.12±8.89	85.23±7.15	**97.41±2.56**
OA	95.17±1.72	94.64±2.73	96.17±1.91	96.83±1.19	95.53±2.18	97.24±1.72	94.60±3.15	97.07±1.41	97.15±1.73	**98.34±1.16**
AA	95.79±2.18	88.53±3.41	93.57±1.19	96.20±2.45	96.33±1.58	95.73±1.14	92.48±2.30	95.85±2.16	90.76±4.51	**98.31±0.57**
Kappa	93.73±3.35	92.98±2.18	94.97±2.72	95.87±2.24	94.20±2.21	97.66±2.03	92.97±1.19	96.17±2.14	96.52±2.33	**97.83±1.26**

TABLE V
QUANTITATIVE RESULT (ACC% ± STD%) OF HU. BEST IN BOLD AND SECOND WITH UNDERLINE

Class	AMGCFN TGRS2023	GSC-ViT TGRS2024	MASSFormer TGRS2024	S^2GFormer TGRS2024	PSFormer TGRS2024	DBCT TGRS2024	GiGCN TNNLS2024	NESSGGCN ESWA2025	SSEFN TGRS2025	SPGFormer Ours
1	88.00±7.14	92.46±6.64	95.22±1.57	94.34±4.62	85.67±3.12	82.84±5.24	84.06±4.98	86.59±5.73	**95.42±2.48**	87.08±3.61
2	92.92±1.16	94.68±5.07	96.12±2.14	**98.06±1.45**	93.70±3.81	79.25±8.18	79.33±6.73	92.98±4.00	68.91±15.04	92.62±2.04
3	99.44±0.16	89.44±18.73	99.85±0.09	99.71±0.33	**100.00±0.00**	98.97±0.35	**100.00±0.00**	99.97±0.06	**100.00±0.00**	**100.00±0.00**
4	89.51±2.31	94.76±2.30	92.35±5.30	98.78±1.10	**100.00±0.00**	94.38±2.33	93.40±4.13	95.04±3.42	99.73±0.32	**100.00±0.00**
5	98.63±1.68	95.91±6.14	**100.00±0.00**	**100.00±0.00**	99.89±0.11	**100.00±0.00**	**100.00±0.00**	96.74±5.00	95.55±3.63	**100.00±0.00**
6	92.25±4.06	94.97±5.52	87.09±4.14	**100.00±0.00**	82.58±10.52	95.48±3.22	93.87±3.12	88.45±5.44	**100.00±0.00**	97.04±1.54
7	90.08±5.52	81.06±8.34	89.22±3.47	83.72±5.37	89.70±5.17	**96.24±1.30**	89.54±3.30	79.58±11.71	92.94±4.16	94.23±2.16
8	57.82±10.99	68.75±8.95	66.47±17.85	70.06±13.37	55.24±10.51	68.42±10.78	66.96±4.06	73.43±7.40	**95.27±1.47**	81.86±3.89
9	78.78±12.90	81.49±5.99	85.69±5.10	76.07±0.00	87.22±3.55	90.05±4.11	81.00±7.67	82.33±6.18	89.03±4.08	**94.07±2.55**
10	91.05±11.08	84.20±12.72	**98.34±0.43**	79.95±10.17	95.04±2.36	94.30±1.92	92.98±5.35	75.54±9.53	77.61±8.55	98.09±2.01
11	93.78±3.26	80.62±12.09	94.34±2.32	97.38±1.16	86.72±4.16	**98.44±0.86**	97.21±1.25	84.80±4.72	83.04±6.98	91.02±5.48
12	78.28±11.36	76.94±10.63	**94.58±2.32**	73.48±12.57	73.15±13.20	86.94±4.05	69.04±6.09	68.93±8.06	94.53±2.24	93.32±3.49
13	92.94±1.15	92.58±5.19	76.87±7.32	**94.27±2.21**	80.61±8.05	91.62±3.57	62.11±6.47	89.56±4.09	80.11±7.06	89.53±2.61
14	**100.00±0.00**	98.01±4.70	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.76±0.05	97.86±1.99	**100.00±0.00**
15	**100.00±0.00**	98.84±1.27	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
OA	87.87±4.35	86.69±2.16	91.71±1.81	89.17±3.99	87.95±2.49	90.52±1.70	86.53±3.15	85.82±2.51	88.95±2.07	**93.98±1.37**
AA	89.57±3.73	88.31±1.71	91.74±2.96	91.05±2.54	88.64±1.95	91.80±1.43	87.30±2.01	87.58±2.18	91.33±2.11	**94.59±1.28**
Kappa	86.90±4.68	85.62±2.33	91.04±2.27	88.29±2.42	86.97±2.64	89.75±3.03	85.43±2.59	84.67±2.71	88.05±2.55	**93.49±1.78**

HU datasets, respectively. AMGCFN integrates multihop local structural information with convolutional features, reaching improved OA values of 94.01%, 95.32%, 95.17%, and 87.87%, validating the importance of local structural information in HSI classification. ViT-based methods demonstrate varying performance across datasets. DBCT's dual-branch structure balances global information capture with local feature extraction, obtaining OA values of 92.18%, 94.43%, 97.24%, and 90.52%. GSC-ViT reduces parameters but shows limited performance in complex boundary regions with OA values of 86.45%, 89.43%, 94.64%, and 86.69%. The fusion method S^2GFormer shows promising results on structurally complex datasets like LK (96.83% OA), proving the potential of integrating graph structure information with global attention mechanisms. Unlike methods that use image patches as processing units, our SPGFormer processes at the superpixel level. Through graph Laplacian PE and structure-aware attention mechanisms, it effectively combines global position information with local topological structure. Our method improves OA values by 2.04%, 2.39%, 1.10%, and 2.27% over second-best methods across the four datasets, with consistently smaller standard deviations (e.g., UP dataset OA standard deviation of only ±1.18%), demonstrating strong stability across different training–testing splits.

Despite the overall superior performance of SPGFormer, it is worth noting that certain challenging classes exhibit relatively lower accuracies. For example, HU-Class 8 and UP-Class 1 show performance gaps compared with the optimal methods. These failures primarily occur in highly heterogeneous urban areas where superpixel segmentation forces diverse land covers into single nodes, resulting in oversimplified representations that lose discriminative details. In addition, spectrally similar classes (e.g., healthy versus stressed grass) pose challenges when boundary regions are incorrectly merged during superpixel generation. These limitations suggest future work should develop more adaptive structure-spatial–spectral modeling strategies to handle complex land-cover distributions and improve model robustness.

2) Visual Analysis of Classification Results: Figs. 7–10 display the classification maps obtained by different methods on the UP, SA, LK, and HU datasets for qualitative evaluation. From the visual analysis, GCN-based methods (such as GiGCN) can capture spatial contextual information but perform poorly when processing complex boundaries and small regions, with some noise still present in the classification maps. ViT-based methods (such as GSC-ViT and MASSFormer) can establish long-range dependencies, improving

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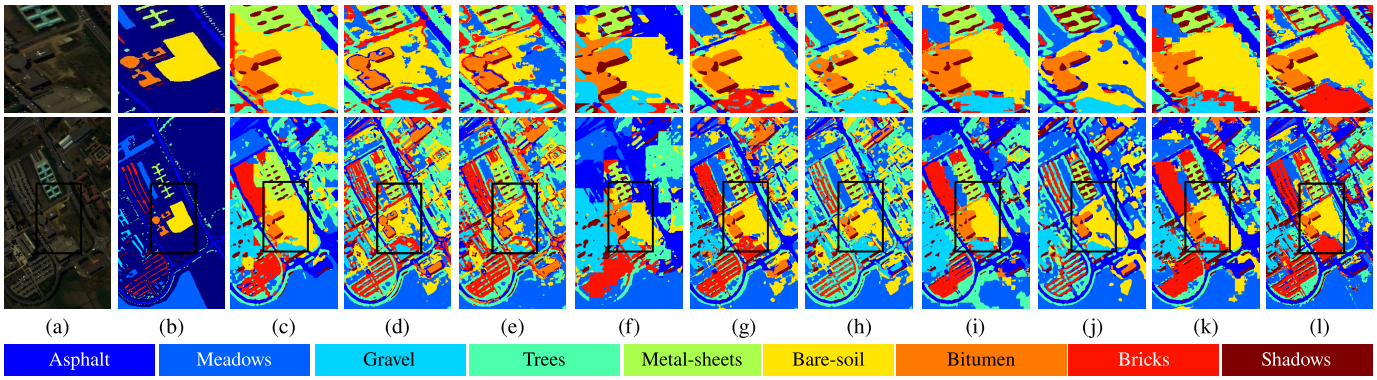


Fig. 7. Classification maps of different methods on the UP dataset. (a) False-color image. (b) Ground truth. (c) AMGCFN. (d) GSC-ViT. (e) MASSFormer. (f) S^{2}GFormer. (g) PSFormer. (h) DBCT. (i) GiGCN. (j) NESSGGCN. (k) SSEFN. (l) SPGFormer.

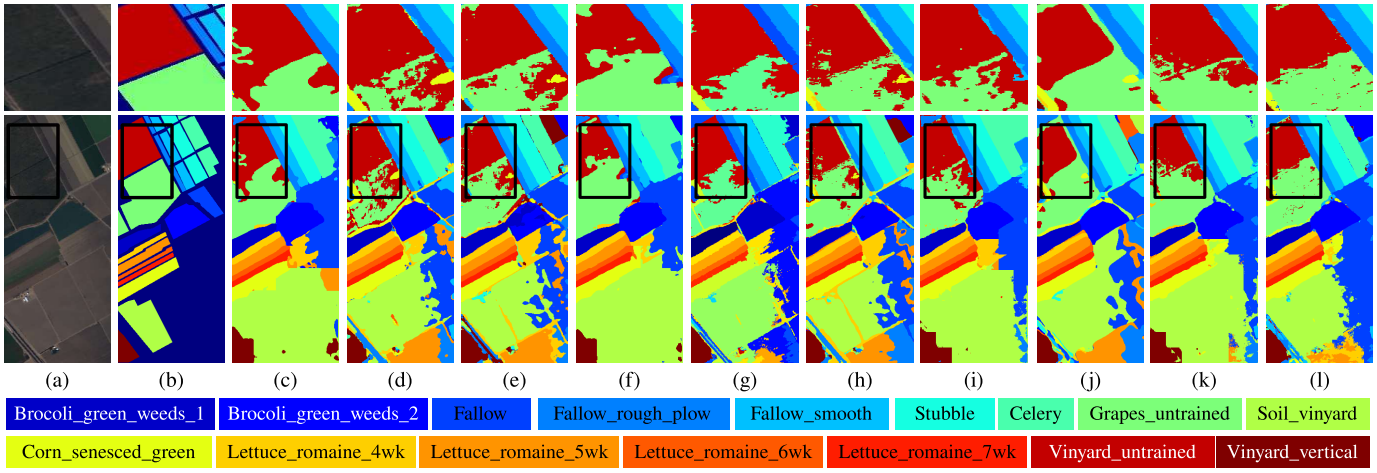


Fig. 8. Classification maps of different methods on the SA dataset. (a) False-color image. (b) Ground truth. (c) AMGCFN. (d) GSC-ViT. (e) MASSFormer. (f) S^{2}GFormer. (g) PSFormer. (h) DBCT. (i) GiGCN. (j) NESSGGCN. (k) SSEFN. (l) SPGFormer.

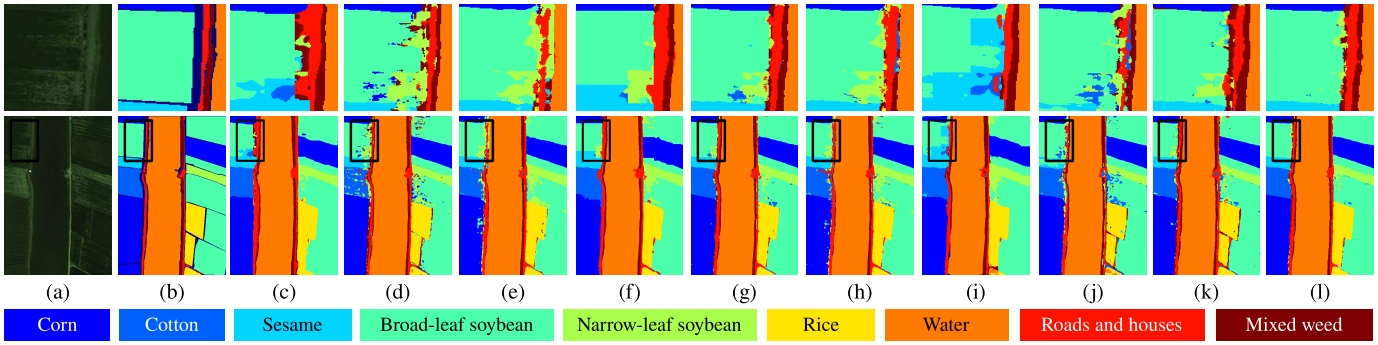


Fig. 9. Classification maps of different methods on the LK dataset. (a) False-color image. (b) Ground truth. (c) AMGCFN. (d) GSC-ViT. (e) MASSFormer. (f) S^{2}GFormer. (g) PSFormer. (h) DBCT. (i) GiGCN. (j) NESSGGCN. (k) SSEFN. (l) SPGFormer.

the over-smoothing problem of edge features to some extent. The classification maps clearly show that SPGFormer produces smoother and more accurate classification results across all datasets. On the UP dataset, SPGFormer maintains consistency in homogeneous regions and clear boundaries in complex areas. On the SA dataset, SPGFormer achieves better classification performance for easily confused regions (such as classes 5 and 8) while maintaining structural integrity. On the LK dataset, SPGFormer maintains the continuity of classification results in complex strip-shaped regions, and in the more challenging HU dataset, SPGFormer accurately classifies numerous small objects and complex structures. Overall, the visual comparison clearly demonstrates that SPGFormer can better represent the boundary and detail characteristics. This qualitative assessment aligns with the quantitative results, further confirming the superiority of our proposed method in HSI classification tasks.

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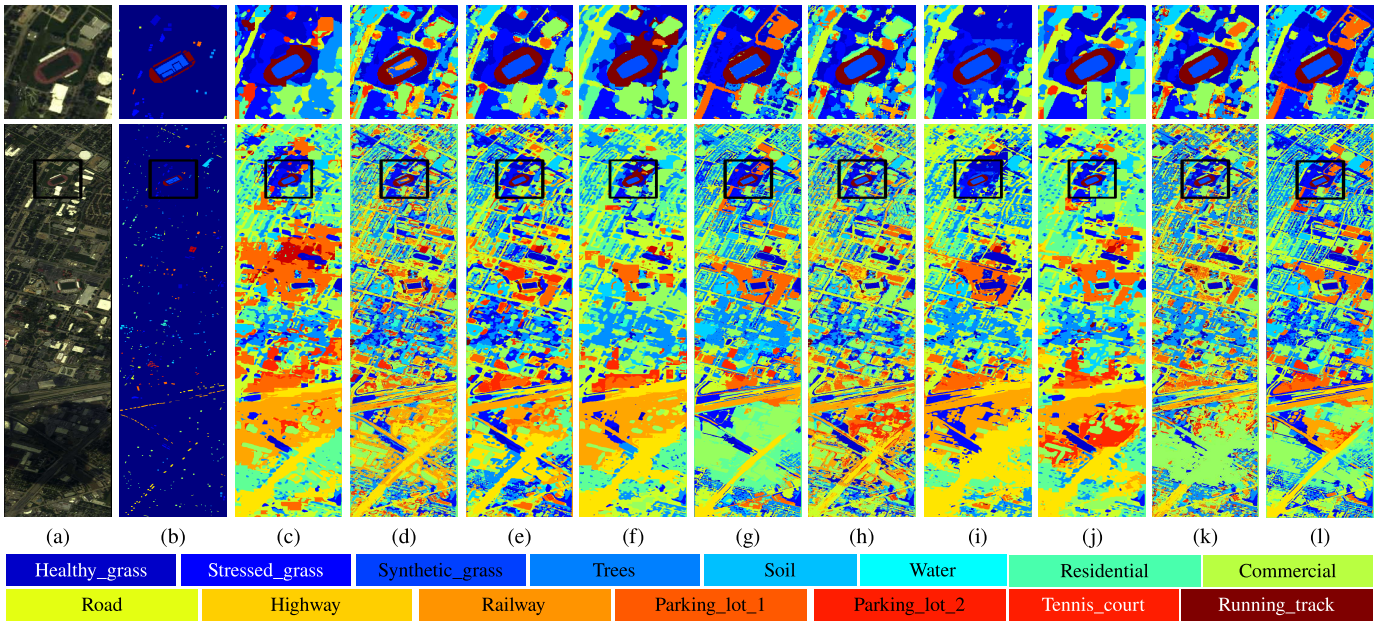


Fig. 10. Classification maps of different methods on the HU dataset. (a) False-color image. (b) Ground truth. (c) AMGCFN. (d) GSC-ViT. (e) MASSFormer. (f) S^{2}GFormer. (g) PSFormer. (h) DBCT. (i) GiGCN. (j) NESSGGCN. (k) SSEFN. (l) SPGFormer.

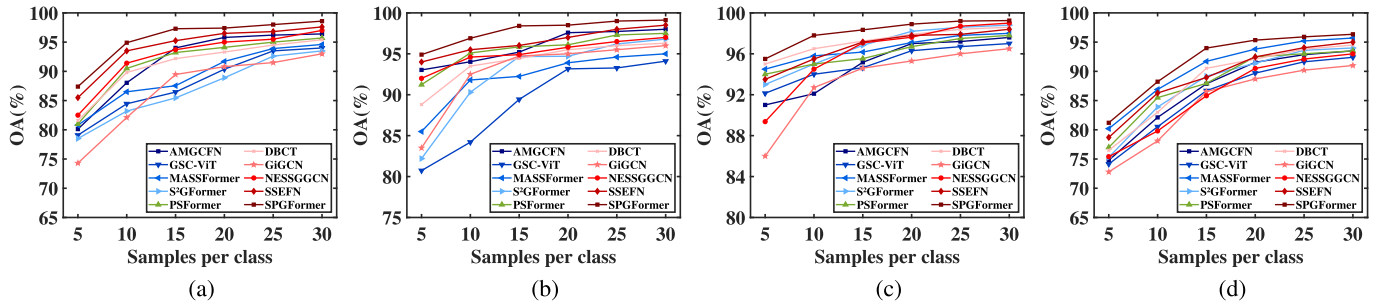


Fig. 11. Classification accuracies of different methods under different numbers of training samples on each dataset. (a) UP. (b) SA. (c) LK. (d) HU.

E. Influence of Training Set Size

In practical applications, obtaining large quantities of labeled training data for HSI classification is often chal lenging and resource-intensive. To evaluate the robustness of our proposed SPGFormer when limited training data are available, we conducted experiments with varying numbers of training samples in the range of {5, 10, 15, 20, 25, 30} per class for the four datasets. It should be noted that the graph construction process utilizes all pixel data from the entire HSI to ensure complete representation of the spatial topological structure, while labeled samples are only used during the model training phase for supervised learning of the classifier parameters. This design fully leverages the semi-supervised characteristics of graph-based methods. Fig. 11 illustrates the classification performance in terms of OA with different sizes of the training set. As expected, all methods demonstrate improved performance as the number of training data increases. However, SPGFormer consistently achieves superior or competitive accuracies across all datasets and training sample settings. On the UP dataset, SPGFormer maintains a significant advantage over other methods, particularly when training samples are extremely limited. With only five samples per class, SPGFormer achieves an OA of 87.39%, outperforming the second-best method SSEFN by 2.11%. This gap narrows slightly, but remains substantial with increasing training samples, with improvements of 1.78%, 2.14%, 1.12%, 1.21%, and 1.05% when using 10, 15, 20, 25, and 30 samples per class, respectively. The SA, LK, and HU datasets follow similar patterns, with SPGFormer maintaining performance advantages, especially in low-sample scenarios.

Notably, when using extremely limited training data (e.g., five samples per class), most Transformer-based methods (such as MASSFormer and GSC-ViT) experience a substantial decrease in accuracy due to their large parameter space and potential for overfitting. In contrast, SPGFormer's dual-branch architecture with graph Laplacian PE enables more efficient learning from limited samples, making it particularly valuable for practical applications where labeled data are scarce.

F. Ablation Experiments

In this section, extensive ablation experiments are conducted on four benchmark datasets to investigate the contributions of the main modules of SPGFormer, including the NFT branch, the SPPT branch, the SPMHSA mechanism, the BCAM,

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TABLE VI
ABLATION ANALYSIS ON FOUR DATASETS. BEST IN BOLD

Cases	NFT	SPPT	SPMHSA	BCAM	LFP	UP	SA	LK	HU
1	–	√	√	–	√	93.43±2.58	95.17±2.76	93.26±3.11	88.25±2.49
2	√	–	–	–	√	94.21±1.91	95.73±1.57	94.85±1.82	90.74±2.12
3	√	√	–	√	√	95.14±0.95	97.17±1.13	96.81±1.51	91.89±1.16
4	√	√	√	–	√	95.86±1.36	96.63±0.87	96.97±1.58	91.54±1.32
5	√	√	√	√	–	96.08±0.54	97.84±0.78	97.56±1.19	92.42±1.28
6	√	√	√	√	√	**97.29±1.18**	**98.41±1.01**	**98.34±1.16**	**93.98±1.37**

and the learnable fusion parameter α (denoted as LFP). The experimental results are shown in Table VI.

Cases 1 and 2 evaluate single-branch performance. Case 1 retains only the SPPT branch (including SPMHSA) to assess structural modeling capability with Laplacian positional encoding. Case 2 retains only the NFT branch and incorporates standard sinusoidal positional encoding, enabling the model to isolate the contribution of Laplacian PE. Note that BCAM is removed in both cases as it requires dual branches for interaction. Cases 3–5 systematically ablate components in the dual-branch architecture. Case 3 substitutes SPMHSA with standard MHSA to verify the importance of structure-aware attention. Case 4 replaces BCAM with simple feature concatenation to quantify the benefit of bidirectional cross attention. Case 5 removes the learnable fusion parameter (fixed $\alpha = 0.5$) to assess adaptive fusion. Case 6 represents the complete model as our performance baseline.

The experimental results reveal several key findings. First, both single-branch models achieve reasonable performance but remain significantly below the complete model, confirming the complementary nature of spectral and structural information. Among the dual-branch variants, removing the SPMHSA mechanism (Case 3) causes notable performance deterioration, with decreases of 2.15%, 1.24%, 1.53%, and 2.09% across the four datasets, validating the importance of incorporating k-hop structural information. Removing the BCAM module (Case 4) similarly leads to significant performance degradation, with drops of 1.43%, 1.78%, 1.37%, and 2.44% across the four datasets, highlighting the critical role of cross-branch interaction. While removing the learnable fusion parameter (Case 5) has a relatively smaller impact, it still results in drops of 1.21%, 0.57%, 0.78%, and 1.56% across the four datasets, demonstrating the value of adaptive feature fusion.

G. Visualization Analysis of Effectiveness of LPE

To empirically validate the effectiveness of Laplacian PE, we conducted t-SNE visualization experiments comparing LPE with two commonly used PE methods, namely, SPE and RWPE. As illustrated in Fig. 12, LPE demonstrates superior clustering properties across all datasets. The visualization results reveal that LPE produces more compact and well-separated clusters, effectively preserving the intrinsic spatial–spectral structure of HSI data.

For the UP dataset, classes 3 (gravel) and 8 (bricks) share similar spectral signatures but are distinctly separated under LPE, while showing significant overlap with SPE and RWPE. Similarly, in the SA dataset, classes 8 (grapes untrained) and 15 (vinyard untrained), both vineyard-related categories with subtle spectral differences, achieve excellent separability

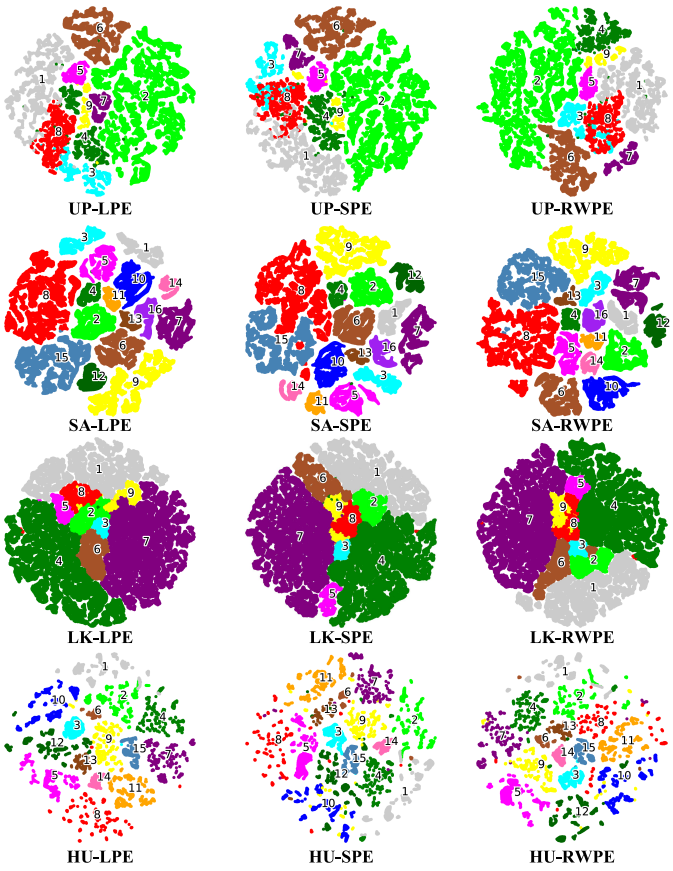


Fig. 12. t-SNE visualization with different PE methods on four HSI datasets.

with LPE. For the LK dataset, which represents a relatively simple agricultural classification task, LPE achieves optimal clustering with minimal interclass confusion, demonstrating its effectiveness even in straightforward scenarios. Most importantly, in the challenging HU dataset with 15 complex urban land-cover types, LPE maintains remarkable separation between classes. These visualizations confirm that Laplacian eigenvectors capture the graph's intrinsic geometry more effectively than alternative PE methods, providing discriminative representations that are crucial for accurate HSI classification across diverse scenarios ranging from agricultural to complex urban environments.

H. Complexity Analysis

Table VII presents computational efficiency metrics for all compared methods. SPGFormer maintains a parameter count of approximately 350k, which remains highly competitive among all methods, being substantially smaller than approaches like SSEFN (6.2M–8.8M), GSC-ViT (543k–578k), and GiGCN (640k–682k). In terms of computational complexity, SPGFormer demonstrates exceptional efficiency by achieving the lowest FLOPs on three datasets and the second-lowest on one dataset (7.15G–28.18G). Training efficiency is equally impressive, with SPGFormer ranking first on three datasets (8.44–25.64 s) and remaining competitive on the fourth. Inference time ranges from 2.35 to 17.27 s, achieving the fastest speed on HU and the second-fastest on SA. These

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TABLE VII
COMPLEXITY ANALYSIS OF ALL METHODS ON FOUR DATASETS. BEST IN BOLD AND SECOND WITH UNDERLINE

Methods	UP				SA				LK				HU			
	Param(K)	FLOPs(G)	Train(s)	Test(s)	Param(K)	FLOPs(G)	Train(s)	Test(s)	Param(K)	FLOPs(G)	Train(s)	Test(s)	Param(K)	FLOPs(G)	Train(s)	Test(s)
AMGCFN	338.27	14.71	45.33	**2.64**	363.66	10.76	27.75	**2.03**	371.33	23.1	50.44	**14.94**	351.01	55.76	89.65	11.71
GSC-ViT	543.45	24.81	81.95	23.59	568.85	38.45	74.92	11.24	578.36	92.36	78.49	27.84	551.33	241.57	244.28	40.75
MASSFormer	304.24	14.89	22.78	5.38	345.21	17.59	31.54	7.64	360.55	25.67	97.16	18.49	334.88	81.46	169.11	35.81
S^2GFormer	366.58	16.87	50.18	12.86	367.99	32.95	82.53	14.58	366.58	67.49	158.62	31.66	367.78	119.46	216.23	34.59
PSFormer	468.72	11.49	82.65	3.67	482.76	7.88	36.26	2.89	495.65	26.58	88.91	16.68	474.83	40.18	176.43	8.85
DBCT	**22.56**	16.55	48.17	3.16	**30.66**	12.78	45.55	4.98	**40.52**	41.89	118.85	31.44	**22.23**	71.16	215.28	12.63
GiGCN	640.55	9.27	62.56	2.84	660.55	8.19	75.96	5.79	682.54	15.43	91.42	15.28	671.52	30.52	104.24	16.48
NESSGGCN	384.62	12.48	56.48	11.25	374.52	11.56	55.84	10.48	386.04	16.27	58.71	16.84	370.05	**26.84**	**67.82**	10.76
SSEFN	6222.53	97.25	263.2	20.45	6555.23	44.51	254.14	32.85	8826.44	155.87	221.23	327.64	7775.52	348.57	449.46	116.37
Proposed	341.12	**7.15**	**14.19**	3.46	353.54	**6.81**	**8.44**	2.35	358.79	**10.42**	**25.64**	17.27	361.57	28.18	134.55	**7.84**

results confirm that SPGFormer successfully balances model complexity with computational efficiency, making it particularly suitable for real-world HSI classification applications where both accuracy and speed are critical.

V. CONCLUSION

This article proposes the SPGFormer model for HSI classification to address the limitations of modeling positional information and complex spatial relationships. The core innovation of SPGFormer lies in its dual-branch Transformer architecture, and the NFT branch focuses on spectral information processing, while the SPPT branch deeply models graph topological structures. By introducing Laplacian PE, the model can more accurately capture positional relationships in graph structures, overcoming the limitations of traditional methods that integrate positional information primarily through simple addition. The design of SPMHSA and BCAM further enhances the model's capability to model multiscale spatial relationships. Experimental results demonstrate that SPGFormer achieves superior classification performance, particularly excelling in scenarios with limited labeled samples. These advantages make it particularly well-suited for real-world applications such as precision agriculture, urban planning, and environmental monitoring, especially in scenarios requiring complex spatial–spectral joint modeling with limited labeled samples. Future work can further explore more efficient PE methods and cross-modal information fusion strategies.

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